Quasi-Monte Carlo: Foundations

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Abstract

We introduce the framework of quasi-Monte Carlo (QMC) integration: deterministic sequences of points designed to fill the unit cube more uniformly than random samples, achieving asymptotic error $O((\log n)^d/n)$ in dimension d versus the standard $O(1/\sqrt{n})$ of Monte Carlo. The price is the loss of statistical machinery (no central limit theorem, no confidence intervals); we recover this via randomized QMC (RQMC) with digital shifting. We develop the discrepancy framework, prove the Koksma-Hlawka inequality, and construct two foundational sequences: Halton (an extension of van der Corput's one-dimensional sequence) and Sobol (based on primitive polynomials over $\mathbb{F}_2$). The relation to inverse sampling — adopted in the Monte Carlo foundations specifically for QMC compatibility — is highlighted throughout.

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1 From Monte Carlo to quasi-Monte Carlo

The Monte Carlo identity is

$$\int_{[0,1]^d} f(u)\,\mathrm{d}u \;\approx\; \frac{1}{n}\sum_{i=1}^{n} f(u_i), \qquad u_i \sim \mathrm{Unif}\big([0,1]^d\big) \text{ i.i.d.}, \tag{1}$$

with statistical error of order $\sigma(f)/\sqrt{n}$ where $\sigma^2(f) = \mathrm{Var}(f(u))$. The variance reduction techniques (antithetic and control variates) attack the constant $\sigma(f)$; they leave the rate $1/\sqrt{n}$ untouched.

QMC takes a different route. Instead of choosing $\{u_i\}$ randomly, we choose them deterministically to be "as uniform as possible" in $[0,1]^d$. The hope is that the average $\frac{1}{n}\sum f(u_i)$ converges to the integral faster than $1/\sqrt{n}$.

The challenge is to make "as uniform as possible" precise. The notion is captured by the *discrepancy* of a point set, which measures the worst-case deviation of the empirical distribution from the uniform distribution on rectangles.

Definition 1.1 (star discrepancy)**.** For a finite point set $P = \{u_1, \ldots, u_n\} \subset [0,1)^d$ and a rectangle $B = \prod_{j=1}^{d}[0, b_j) \subset [0,1)^d$, let

$$\mathrm{D}^*(P; B) \;=\; \left| \frac{\#\{i : u_i \in B\}}{n} - \mathrm{vol}(B) \right|. \tag{2}$$

The *star discrepancy* of P is

$$\mathrm{D}^*(P) \;=\; \sup_{B \subset [0,1)^d} \mathrm{D}^*(P; B). \tag{3}$$

A sequence with small discrepancy distributes points evenly across all anchored rectangles. For random points, $\mathbb{E}[\mathrm{D}^*(P)] = \Theta(\sqrt{(\log\log n)/n})$ by the law of the iterated logarithm; for QMC sequences, deterministic constructions can achieve $\mathrm{D}^*(P) = O((\log n)^d/n)$.

Theorem 1.2 (Koksma-Hlawka inequality)**.** *For any function $f : [0,1]^d \to \mathbb{R}$ of bounded variation in the sense of Hardy-Krause, with variation $V(f)$, and for any finite point set $P \subset [0,1)^d$,*

$$\left| \frac{1}{n}\sum_{i=1}^{n} f(u_i) \;-\; \int_{[0,1]^d} f(u)\,\mathrm{d}u \right| \;\le\; V(f) \cdot \mathrm{D}^*(P). \tag{4}$$

The Koksma-Hlawka bound is *deterministic*: no probability is involved. Its strength is that, for nice f (smooth, bounded mixed partial derivatives), $V(f) < \infty$, and the integration error decays as fast as the discrepancy. Its weakness, on which we return below, is that $V(f)$ is typically very hard to bound (and often infinite for the indicators that appear in option payoffs), so the inequality gives no usable a priori error estimate.

1.1 The asymptotic comparison

For Halton or Sobol sequences in dimension d, the discrepancy satisfies $\mathrm{D}^*(P_n) = O((\log n)^d/n)$. Compared to MC:

Method	Error rate	Behaviour as $n \to \infty$
MC	$O(n^{-1/2})$	always wins eventually
QMC, d moderate	$O((\log n)^d/n)$	wins MC for moderate n
QMC, d large	$O((\log n)^d/n)$	loses to MC for practical n

The crossover point depends on d. For $d \le 5$, QMC dominates MC at any $n \ge 10^3$. For $d = 30$, the $(\log n)^d$ factor is catastrophic — $(\log 10^6)^{30} \approx 10^{34}$ — and QMC requires extreme n to outperform MC. In practice, QMC is most useful for low-to-moderate dimensional problems.

A modern viewpoint, developed by Sloan, Woźniakowski, Owen and others, refines this picture: many financial integrands, despite high nominal dimension, have "effective dimension" much smaller than d because the function depends primarily on a low-dimensional subspace. QMC then performs well even at large nominal d. This is beyond the scope of the present writeup but motivates the preparatory experiments of the QMC pricer for the European call.

2 The van der Corput sequence

The simplest QMC construction is one-dimensional. Given an integer base $b \geq 2$, write each non-negative integer i in base b as

$$i \;=\; \sum_{k\geq 0} a_k(i)\, b^k, \qquad a_k(i) \in \{0, 1, \ldots, b-1\}. \tag{5}$$

The *radical inverse function* $\phi_b : \mathbb{N} \to [0, 1)$ reverses the digits and places a decimal point:

$$\phi_b(i) \;=\; \sum_{k\geq 0} a_k(i)\, b^{-(k+1)}. \tag{6}$$

The *van der Corput sequence in base* b is $\{\phi_b(i)\}_{i\geq 1}$.

Example 2.1 (base 2). The base-2 van der Corput sequence starts

$$\tfrac{1}{2},\ \tfrac{1}{4},\ \tfrac{3}{4},\ \tfrac{1}{8},\ \tfrac{5}{8},\ \tfrac{3}{8},\ \tfrac{7}{8},\ \tfrac{1}{16},\ \tfrac{9}{16}, \ldots$$

After 2^k points, the sequence has used each "odd numerator over 2^{k+1}" exactly once, so the points are perfectly equidistributed in $[0, 1)$ at scale $2^{-(k+1)}$.

The van der Corput sequence in base b has discrepancy $\mathrm{D}^* = O(\log n/n)$, optimal in $d = 1$.

3 The Halton sequence

The natural multi-dimensional extension is to use a different prime base in each coordinate. Choose primes $p_1 < p_2 < \cdots < p_d$ (e.g., $p_1 = 2, p_2 = 3, p_3 = 5, \ldots$), and define the Halton sequence by

$$\mathbf{u}_i \;=\; \big(\phi_{p_1}(i), \phi_{p_2}(i), \ldots, \phi_{p_d}(i)\big) \in [0, 1)^d. \tag{7}$$

The construction is constructive, infinite, and trivial to implement: for each new index i, evaluate d radical inverses.

The Halton sequence has discrepancy $\mathrm{D}^* = O((\log n)^d/n)$. The constant in the big-O grows with d: more precisely, the asymptotic constant is $\prod_{j=1}^{d} \frac{p_j - 1}{2 \log p_j}$, which for p_d around 20 (i.e., $d \approx 8$) becomes uncomfortably large.

The high-dimensional pathology. Halton's reliance on primes that grow with dimension introduces a well-known artefact: for $d \geq 7$, projections of the Halton sequence onto pairs of high-index dimensions exhibit visible linear correlations. The prime $p_8 = 19$ is large enough that the radical inverse "advances" very slowly within a window of consecutive indices, producing alignment with $p_9 = 23$. This is why Sobol, despite being more complex to construct, is preferred in practice.

Implementation. The radical inverse is computed iteratively: while $i > 0$, extract the lowest digit of i in base b, multiply by $b^{-(k+1)}$, accumulate. The whole Halton sequence in dimension d requires $O(d \log_2 n)$ digit operations per point.

4 The Sobol sequence

Sobol sequences are the standard choice for QMC in finance. Whereas Halton uses a different prime base per dimension, Sobol uses base 2 in every dimension and varies a different ingredient: the *direction numbers*, which encode primitive polynomials over $\mathbb{F}_2$.

4.1 Direction numbers and the recursion

For each dimension j, fix a sequence of *direction numbers* $v_1^{(j)}, v_2^{(j)}, \ldots$ where each $v_k^{(j)} \in (0, 1)$. The Sobol value for index i in dimension j is

$$u_i^{(j)} \;=\; b_1(i)\, v_1^{(j)} \;\oplus\; b_2(i)\, v_2^{(j)} \;\oplus\; \cdots \;\oplus\; b_K(i)\, v_K^{(j)}, \tag{8}$$

where $b_k(i)$ are the binary digits of the index i ($i = \sum_k b_k(i) 2^{k-1}$), K is the number of bits needed to represent i, and $\oplus$ is bitwise XOR (interpreted as addition modulo 2 on the binary expansion of $v_k^{(j)}$).

For the construction to produce a low-discrepancy sequence, the direction numbers must satisfy specific algebraic properties. They are derived from a primitive polynomial over $\mathbb{F}_2$ for each dimension; the construction is due to Sobol (1967) and refined by Joe and Kuo (2008) into a tabulated form.

4.2 The Joe-Kuo tables

Joe and Kuo published direction-number tables for dimensions up to 21 201 in their 2008 paper. The tables are public and freely distributed at the URL web.maths.unsw.edu.au/∼fkuo/sobol/. They are the standard reference used by virtually all production implementations of Sobol in finance (including `scipy.stats.qmc.Sobol`, which loads them at import time).

The structure of the table is, for each dimension d:

- A primitive polynomial degree s and an integer encoding a representing the polynomial coefficients.
- Initial direction numbers $m_1, m_2, \ldots, m_s$ (one per degree-of-freedom of the polynomial).

The remaining direction numbers are generated by the recursion

$$m_k \;=\; 2a_1 m_{k-1} \oplus 2^2 a_2 m_{k-2} \oplus \cdots \oplus 2^{s-1} a_{s-1} m_{k-s+1} \oplus (2^s m_{k-s} \oplus m_{k-s}), \quad k > s, \tag{9}$$

where a_j are the polynomial coefficients (with $a_0 = a_s = 1$ by convention). The direction numbers $v_k = m_k / 2^k$ are then inserted in equation (8).

The construction may look intimidating; in practice, it is ~ 50 lines of code once the Joe-Kuo file is loaded.

4.3 The Gray-code optimisation

A naive implementation evaluates the XOR over K direction numbers for each new index i. Sobol observed that consecutive indices in *Gray-code order* differ in exactly one bit, so generating u_{i+1} from u_i requires only one XOR. This optimisation is essential for performance; we adopt it in the C++ implementation.

5 Quasi-Monte Carlo with inverse sampling

Suppose we want to estimate $\mathbb{E}[g(Z)]$ where $Z \sim \mathcal{N}(0, 1)^d$. The MC route is:

1. Draw $u_i \sim \mathrm{Unif}([0, 1)^d)$ i.i.d.
2. Apply $z_i = \Phi^{-1}(u_i)$ componentwise.
3. Average $g(z_i)$.

The QMC route replaces step 1 by a low-discrepancy sequence and keeps steps 2-3 unchanged. *Crucially, this only works if the mapping Φ^{-1} is monotone and one-to-one.* Other normal samplers — Box-Muller (which mixes pairs of uniforms in a non-monotone way), Marsaglia polar (which involves rejection), Ziggurat (rejection-based) — destroy the low-discrepancy structure of the input. Box-Muller, in particular, can introduce spurious correlations between dimensions when paired with a QMC source.

This is why the Monte Carlo foundations adopt inversion sampling, even though Ziggurat is faster for plain MC. The pipeline

$$\text{QMC source} \longrightarrow \Phi^{-1} \longrightarrow \text{normal samples} \tag{10}$$

preserves the discrepancy structure across the transformation. The choice of normal sampler made in the Monte Carlo foundations is deliberately taken with this need in mind.

6 The need for randomization: from QMC to RQMC

Consider the situation after running a QMC integrator. We have computed

$$\hat{I}_n^{\mathrm{QMC}} \;=\; \frac{1}{n}\sum_{i=1}^{n} f(u_i), \tag{11}$$

a single deterministic number. The Koksma-Hlawka bound gives a deterministic error estimate $|I - \hat{I}_n^{\mathrm{QMC}}| \le V(f)\,\mathrm{D}^*$, but $V(f)$ is essentially never available: the Hardy-Krause variation of an option payoff is hard to bound, often infinite for indicator functions, and not available in closed form even for the European call.

We have a number with no error bar. This is unacceptable in production: pricing requires a quantifiable uncertainty.

6.1 Digital shifting

The fix is to randomise the QMC sequence in a way that preserves its low-discrepancy property but makes each marginal point uniform. The simplest randomisation is the *digital shift*.

Definition 6.1 (digital shift)**.** Given a base-2 QMC sequence $\{u_i\}$ in $[0,1)^d$ and a uniform random shift $\xi \in [0,1)^d$, the digitally shifted sequence is

$$\tilde{u}_i^{(j)} \;=\; u_i^{(j)} \oplus \xi^{(j)} \quad \text{(bitwise XOR)}. \tag{12}$$

For each fixed i and j, $\tilde{u}_i^{(j)} \sim \mathrm{Unif}([0,1))$ marginally; the dependence structure between indices is preserved.

The digital shift is reversible (XOR with ξ twice gives back the original), preserves the local properties of Sobol points (by linearity of XOR over $\mathbb{F}_2$), and makes each individual $\tilde{u}_i$ marginally uniform.

6.2 The RQMC estimator

Definition 6.2 (RQMC estimator with R replications)**.** Choose $R \ge 2$ replications. For each replication $r = 1, \ldots, R$, draw an independent random shift $\xi^{(r)}$ and compute

$$\hat{I}_n^{(r)} \;=\; \frac{1}{n}\sum_{i=1}^{n} f(u_i \oplus \xi^{(r)}). \tag{13}$$

The RQMC estimator is the average across replications,

$$\hat{I}_{n,R}^{\mathrm{RQMC}} \;=\; \frac{1}{R}\sum_{r=1}^{R} \hat{I}_n^{(r)}. \tag{14}$$

The R replication estimates $\{\hat{I}_n^{(r)}\}_{r=1}^R$ are i.i.d., so we have a CLT for them. The RQMC estimator's variance is σ_{rep}^2/R where $\sigma_{\text{rep}}^2 = \text{Var}(\hat{I}_n^{(r)})$, and the half-width is

$$h_{n,R}^{\text{RQMC}} \;=\; z_{\beta/2} \cdot \frac{\hat{\sigma}_{\text{rep}}}{\sqrt{R}}, \tag{15}$$

identical in form to the MC half-width but with the variance estimated from R replication-estimates instead of from n payoffs.

The crucial empirical fact: σ_{rep}^2 scales with the QMC error, so for nice integrands,

$$\sigma_{\text{rep}}^2 \;=\; O\big((\log n)^{2d}/n^2\big) \quad \text{instead of} \quad O(\sigma^2(f)/n). \tag{16}$$

At fixed total budget nR, RQMC produces estimator variance asymptotically smaller than MC by orders of magnitude (for nice f and moderate d).

6.3 The trade-off in choosing n vs R

For a fixed total budget $N = nR$ payoff evaluations:

- Larger n exploits the QMC convergence rate.
- Larger R gives more replications and a tighter half-width estimate (improved by $1/\sqrt{R}$ in the CLT sense, but also useful for stability of $\hat{\sigma}_{\text{rep}}$).

The standard practical rule is $R \in [10, 30]$ replications and the remaining budget on n. We adopt $R = 20$ in the validation script of the QMC pricer for the European call.

7 Summary and look ahead to the QMC pricer

We have established the framework:

- QMC sequences (Halton, Sobol) achieve discrepancy $O((\log n)^d/n)$, faster than the MC rate $O(\sqrt{(\log\log n)/n})$.
- The Koksma-Hlawka bound is deterministic but practically unusable for option payoffs because $V(f)$ is hard to bound.
- Inverse sampling (chosen in the Monte Carlo foundations) is the only normal sampler that preserves QMC structure.
- RQMC with digital shifting recovers a usable error bar by averaging R independent shifts.

The note on the QMC pricer (`mc_european_qmc.tex`) specialises this framework to the European call. It implements Halton and Sobol generators, builds a multi-step Euler MC pricer with $d = 20$ (one normal per discretisation step), compares deterministic-QMC and RQMC against the IID Euler–Maruyama baseline, and measures empirical convergence rates on a log-log slope plot.

8 Bibliographic notes

The standard reference for QMC integration is Niederreiter's monograph [4]. The original Sobol sequence is from [6]. The Joe-Kuo tables and primitive polynomials construction are in [2]; the table file `new-joe-kuo-6.21201` is hosted at [3]. Owen's work on RQMC, including the more sophisticated nested scrambling, is reviewed in [5]. The financial-applications viewpoint, including effective dimension, is developed by Glasserman [1, Chapter 5].

References

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